CLE-105 (Autumn 2025) Assignment-2

1. I am sure all of you have heard of the **Pole Star**. What is its relation to the **Celestial Sphere**? I make the claim that, in principle, there can be **two pole stars**? Do you agree with the statement? Justify your answer in one or two sentences.

2. A vector $\vec{A} = A_1\hat{x} + A_2\hat{y} + A_3\hat{z}$, can also be represented as a column matrix

$$|A\rangle \equiv \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix}.$$

Its **conjugate vector** $\langle A|$ is represented by a row matrix

$$\langle A| \equiv \begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix}.$$

The **inner product** of two vectors $\vec{A}$ and $\vec{B}$ is written as

$$\langle A|B\rangle \equiv \begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix} \equiv \vec{A} \cdot \vec{B}.$$

The **outer product** of two vectors $\vec{A}$ and $\vec{B}$ is written as

$$|A\rangle\langle B| \equiv \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} \begin{bmatrix} B_1 & B_2 & B_3 \end{bmatrix}.$$

Carrying out the multiplication of the column matrix with the row matrix, you get a 3×3 matrix. The outer product of two vectors is also called the **tensor product** of the two vectors.

Question: The **dot product** and the **cross product** you are familiar with are hidden within the *tensor product*. Given the tensor product of two vectors, that is given the 3×3 matrix, define the procedures by which we can obtain the dot product and the cross product of the two vectors.